

Rayyan Hisham

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EDUCATION

California State Polytechnic University, Pomona

B.S. Mechanical Engineering

Pomona, CA

2024 – Present

Fullerton College

A.S. Engineering, Physics, Mathematics

Fullerton, CA

2022 – 2024

PROJECTS & EXPERIENCE

ASSA ABLOY Americas Inc. | Electronic Locks — Mechanical Engineering Intern

May 2026 – Present

- **Prototype Assembly:** Assembled electronic lock prototypes to complete cyclic/load/impact testing to verify adherence with ANSI/BHMA residential lock security standards.
- **Design Validation:** Completed post-test analysis on functionality of electronic lock internals; utilized **Creo** to make design modifications to resolve condensation failures based on testing data.
- **Supplier Qualification:** Reviewed **First Article Inspections (FAI)** reports to verify part compliance for new-supplier transition; identified critical dimensional discrepancies overlooked during initial supplier FAI reports.

Bronco Motorsports | CPP Formula SAE - Vehicle Dynamics Steering Captain

Jun 2025 – Present

- **Inboard Suspension Research:** Designed an inboard suspension system, optimizing damper, rocker, and pull-rod geometry to achieve target motion ratio and progressive wheel-rate vs. heave response; lowered component **CG** by 10%.
- **Steering System Design:** Designed adjustable toe-links and steering column to minimize compliance, eliminate binding, and reduce component weight; optimized steering spatial angle by 10° and engineered a 2x single U-joint architecture, achieving a 21% total weight reduction.
- **Testing Validation:** Developed MATLAB simulation script to model vehicle toe adjustments based on measured vehicle parameters, while reducing setup times between dynamics events, with more accurate & symmetrical toe changes.
- **Modeling Analysis:** Created SolidWorks CAD models with GD&T drawings for manufacturing; performed FEA validation using FEMAP Nastran to verify a minimum safety factor of 1.5, ensuring design viability under calculated maximum loads.
- **Shop & Machining Experience:** Fabricated and assembled vehicle components using manual machining tools, fixtures, and shop equipment.

Ice Cream Scooping Robot | 5-Axis Robotic Arm — Design Engineer Lead

May 2024 – Present

- **CAD Proficiency:** Designed a 5-axis robotic assembly in **SolidWorks**; implemented a belt-driven arm, reducing required torque by up to 90% compared to direct-drive designs.
- **Project Documentation:** Updated **BOMs** using **Excel**; wrote design and stress analysis documentation; created a presentation pitch for project funding.
- **Fabrication & Assembly:** Machined and fabricated components from CAD drawings; assembled robotic prototypes to specification.

SEMA Engine Build Competition — Team Captain

Sep 2018 – Mar 2020

- **High-Speed Rebuilds:** Led a competitive team completing 250+ timed engine rebuilds with precise torque specifications.
- **Diagnostics & Workflows:** Strengthened proficiency in **engine diagnostics** and assembly while executing under time constraints.
- **Technical Proficiency:** Developed a strong foundation in mechanical **hand tools** and internal combustion engines.